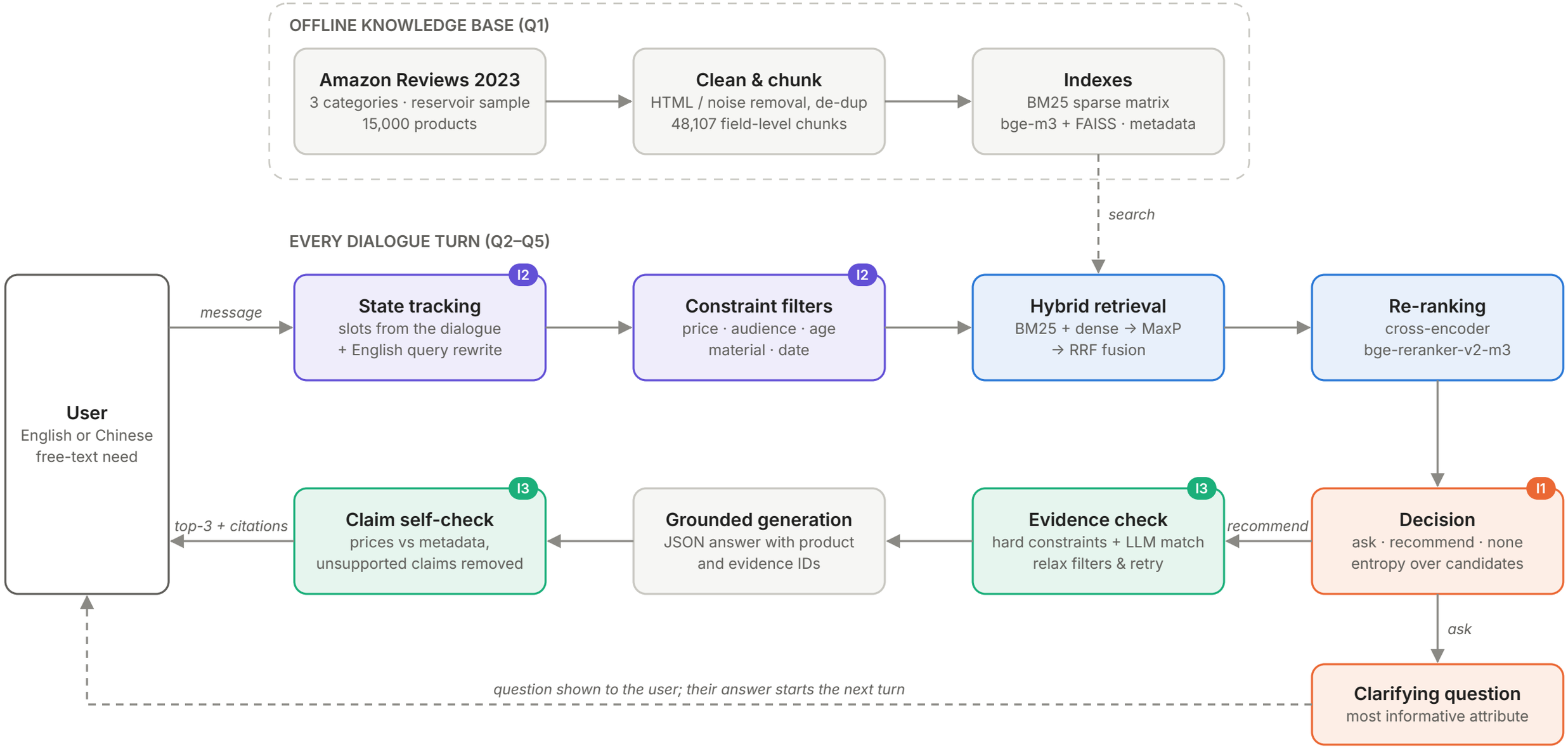
OFFLINE KNOWLEDGE BASE (Q1)
Amazon Reviews 2023
3 categories · reservoir sample
15,000 products
Clean & chunk
HTML / noise removal, de-dup
48,107 field-level chunks
Indexes
BM25 sparse matrix
bge-m3 + FAISS · metadata
search
EVERY DIALOGUE TURN (Q2–Q5)
User
English or Chinese
free-text need
message
I2
State tracking
slots from the dialogue
+ English query rewrite
I2
Constraint filters
price · audience · age
material · date
Hybrid retrieval
BM25 + dense → MaxP
→ RRF fusion
Re-ranking
cross-encoder
bge-reranker-v2-m3
I1
Decision
ask · recommend · none
entropy over candidates
recommend
I3
Evidence check
hard constraints + LLM match
relax filters & retry
Grounded generation
JSON answer with product
and evidence IDs
I3
Claim self-check
prices vs metadata,
unsupported claims removed
top-3 + citations
ask
Clarifying question
most informative attribute
question shown to the user; their answer starts the next turn
query understanding (I2)
retrieval (Q2/Q3)
clarification (I1)
grounding checks (I3)
I1–I3: downstream innovations (Q5)